CERAMIC NOZZLES:

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CAN AN AI AGENT CONTROLLING CAD IMPROVE A BASIC MOULD UNTIL IT CASTS AN ACCURATE CERAMIC ROCKET NOZZLE?

DAYLESFORD COLLEGE
shape the future

RESULT: ~$300 GRAPHITE → $4.50 CERAMIC | BEST SCORE OUT OF 10: 6 (V1) → 9 (V14) | 14 MOULDS, 12 AI-DEVELOPED

HOW IT STARTED:

The team's first quote for a full graphite nozzle showed how expensive it would be. The team has made one, but when lots of them are needed, the cost goes up very quickly. A search for heat-resistant materials showed ceramic near the top of the list. Ceramic could be moulded and fired in the team's kiln. Insanely cheap, durable, no machining, less time and money per nozzle, and a lower barrier to entry for others to try the same thing at a really affordable price.

INTRODUCTION:

Claude Opus 5.5 (an AI agent) operated Onshape to develop the team's basic starting mould (v0), one version at a time. v1-v2 were cast in air-dry clay, v3-v11 were test-fitted only, and v12 onwards were cast in Duracast.

Aim: To test whether an AI agent controlling CAD can improve a basic mould until it casts an accurate, complete Duracast nozzle.

Independent variable: Mould version.

Dependent variables: Score /10 (converging and diverging), human fixes, final throat vs 16 mm.

Controlled variables: Nozzle geometry, release, drying, printer, PLA.

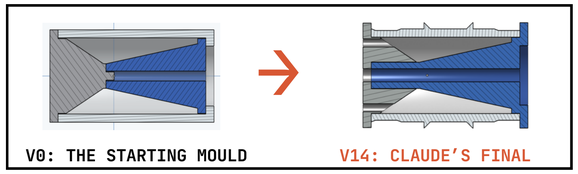
HYPOTHESIS:

If Claude Opus 5.5 (an AI agent) redesigns the mould in Onshape after each test-fit or cast using the team's feedback, then later versions will score higher for both the converging and diverging sections and need fewer human fixes, because each version corrects the last one's faults.

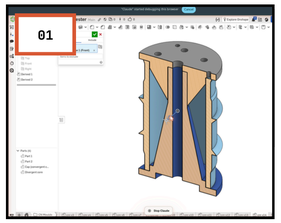
RISKS:

- Refractory dust is harmful to breathe in - wear a P2 mask when mixing
- Wet castable is caustic - gloves and safety glasses
- Burns from the kiln and hot nozzles
- Trapped moisture can flash to steam and burst the nozzle - dry fully before heating
- Nozzle can crack from thermal shock and throw fragments - test remotely behind a barrier

METHOD:



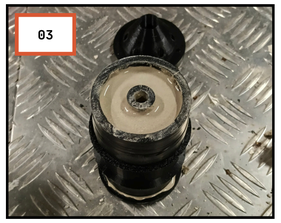
V0: THE STARTING MOULD — V14: CLAUDE'S FINAL



Design a basic mould (v0); Claude develops it in Onshape.



3D print the PLA mould and coat it with release.



Pour the clay or Duracast, demould, then fire Duracast in the kiln.



Test-fit or cast, then score each section /10; the team or Claude fixes the mould.

MATERIALS & SCORING:

- Clay (v1-v2), Duracast (v12+), PLA/PETG printer, release, scales, kiln
- Score /10 per section: 3 checks (0-3) + 1 if throat & exit within 0.5 mm
- Throat & cone angles: custom measuring rod
- Air bubbles: real mass vs mass from density × volume
- Layer lines & cracks: judged by eye

RESULTS:

HUMAN FIXES — CHANGES THE TEAM MADE BY HAND: **12**

THROAT ERROR — FINAL NOZZLE VS 16 MM: **±0.5MM**

COST PER NOZZLE — CERAMIC VS ~$300 GRAPHITE: **~$4.50**

FIGURE 1: SCORES /10 AND HUMAN FIXES BY VERSION (V3-V11 TEST-FIT ONLY, NOT CAST - SEE TABLE 1)

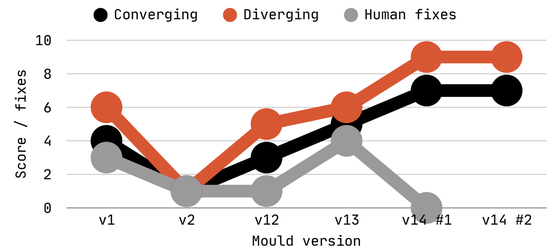






FIGURE 2: BEST NOZZLE (L), SPLIT CAST (R)

Version	Material	Converging /10	Diverging /10	Human fixes	Notes
v0	-	Not cast	Not cast	0	Starting design
v1	Air-dry clay	4	6	3	Close to accurate
v2	Air-dry clay	1	1	1	Clay design; failed
v3-v11	-	Not cast	Not cast	3	Test-fit only
v12	Duracast	3	5	1	Throat off
v13	Duracast	5	6	4	Throat fixed
v14 #1	Duracast	7	9	0	0% rocks, 40 g water, PLA
v14 #2	Duracast	7	9	0	20% rocks, 40 g water, PLA

TABLE 1: MATERIAL, SCORES AND HUMAN FIXES BY VERSION

DISCUSSION:

Mould material: cooled PLA released easily; PETG bonded to the cast and broke the nozzle.

Mix: disk break tests showed 0% rocks was stronger than 20% (40 g water). Both v14 nozzles scored 7 and 9, so the mould cast consistently.

Score dips: v2 and v12 scored below v1. From v2, Claude took steps back and got confused; test-fits showed v3-v11 would do worse, so they were not cast.

Errors & improvements: mostly one nozzle per version; cracks judged by eye. Next: 3 casts per version, 2 scorers, log fixes live.

CERAMIC VS OTHER NOZZLES:

Graphite must be machined (~$300), steel is heavy and erodes, phenolics burn away, copper and tungsten cost far more. Cast ceramic: ~$4.50, no machining, but brittle.

CONCLUSION:

An AI agent controlling Onshape developed the team's basic mould into a v14 mould that cast a complete Duracast nozzle. Scores rose from 4/10 and 6/10 (v1) to 7/10 and 9/10 (v14), and the team made 12 hand fixes, peaking at 4 on v13, so the hypothesis was partly supported. Casting removed all machining and cut nozzle cost from ~$300 to ~$4.50. Converging scores stayed lower because the B1M nozzle's converging section tapers to a thin point that ceramic cannot form, so instead of fixing v14, the team is redesigning the nozzle for ceramic and running new B1M simulations.



FIGURE 3: THE TEAM'S B1M ENGINE, FITTED WITH THE MACHINED GRAPHITE NOZZLE

ROADMAP:

01 NOW: CAST IN AI-DEVELOPED MOULDS (THIS POSTER)

02 NEXT: REDESIGN NOZZLE + NEW B1M SIMULATIONS

03 THEN: STATIC FIRE TEST THE NEW NOZZLE

04 THEN: INTEGRATE INTO B1M + LAUNCH STRAVOX

REFERENCES & ACKNOWLEDGEMENTS:

Anthropic (2026) Claude Opus 5.5 [AI model]. Available at: claude.ai. Artisan Supplies (n.d.) Duracast Castable Refractory 25kg. Available at: artisansupplies.com.au. Graphite Australia (2026) Quote for a machined graphite rocket nozzle [personal communication]. PTC (2026) Onshape [CAD software]. Available at: onshape.com.
Claude designed the moulds in Onshape. Thanks to the Creswick Railway Workshops Association for letting us use their kiln's and FED Uni for making the test hardware.